

AP 2016
Grade-10
General Science Paper-I
Parts A and B

[Time: 2 hrs. 45 min]

[Maximum Marks: 50]

Instructions:

1. This paper contains Part-A
2. Answer the questions. under Part-A on the separate answer book. Answer all the questions: 'Internal choice is given to the questions under Section-III.
3. In the duration, of 2.45 hours, 15 minutes of time is allotted to read the question paper.

Part- A

Time: 2 Hours

Marks: 35

SECTION-I

5x2=10

NOTE:

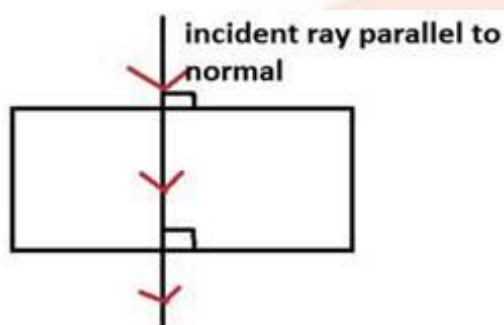
1. Answer ANY FIVE questions, choosing at least TWO from each Group:
2. Each question carries TWO mark.

GROUP-A

1. In what cases, does a light ray not deviate at the interface of two media?

Ans: Normally there are 2 cases in which a ray of light doesn't deviate from the interface of two media:

Case 1: When incidence ray strikes normally at the point of incidence.



Case 2: If the refractive indices of two media are equal.

That is $n_1 = n_2 = n$.

2. What happens to the water when wet clothes dry?

Ans: Due to evaporation of water molecules wet cloths gets dried up. Since due to heat of sunlight the water evaporates and thus causes the cloths to dry.

3. Explain briefly the reason for the blue colour of the sky.

Ans: When the white light of sun passes via atmosphere, the blue colour is scattered in all directions due to the presence of air molecules, that's why most of the time we observe sky as blue.

4. Give any two applications of Faraday's law of Induction in daily life.

Ans: The Faraday's law states that the voltage induced in a coil is directly proportional to rate of change of

electromagnetic flux. **Motors** and **AC (alternating current) generators** are the application of Faraday's law of induction in daily life. Both employs coils of wire and magnetic field to induce voltage.

GROUP-B

5. Why pure acetic acid does not conduct electricity?

Ans: Pure acetic acid doesn't conduct electricity since it doesn't have water molecules so it doesn't dissociate hydronium ions.

6. What is nl^x method? How it is useful?

Ans: It is defined as electronic configuration.

Useful –

- It determines valence electrons of the atom.

It consists of the principal quantum energy level (n) sublevel (l) and the no. of electron (x) in the sublevel. It is useful because, it helps to predict the values of all the four quantum number of any electron.

7. How does metallic character change when we move

(i) Across a period from left to right, (ii) down a group?

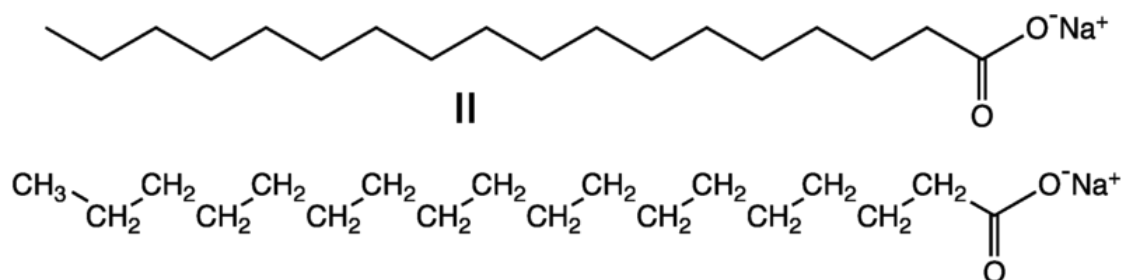
Ans:

(i). Metallic Character decreases from left to right.

(ii). Metallic character increases down the group.

8. Draw the simple figure of a soap molecule.

Ans:



SECTION-II

4x1=4

NOTE:

1. Answer ANY FOUR question from the following.
2. Each question carries ONE marks.

9. Define Latent Heat of Fusion.

Ans: The amount of heat energy released or absorbed when a substance change its form. It could be from liquid to gas or liquid to solid.

10. What is the relationship between focal length (f) and radius of curvature (R)?

Ans: The relationship between focal length (f) and radius of curvature (R) is-

$$r=2f$$

11. What is electric shock?

Ans: When the current flows via a human body and the path shows limited resistance then a body gets an electric shock.

12. Why do we apply paint on iron articles?

Ans: Since paint act as a preventive coating on iron articles so its surface doesn't come in direct connect with air and thus preventing it from rusting.

13. Which group elements are called Carbon family?

Ans: Group 14 elements are called Carbon family.

14. Define Isomerism.

Ans: When a compound has a same chemical formula but different chemical structure is known as isomerism. This is due to various structural and spatial arrangements.

SECTION-III

4x4=16

NOTE:

1. Answer ANY FOUR questions from the following, choosing at least TWO from each group A and B.

2. Each question carries FOUR marks.

GROUP-A

15. Answer these:

- (a) How much energy is transferred when 1 gm of boiling water at 100°C condenses to water at 100°C ?
- (b) How much energy is transferred when 1 gm of boiling water at 100°C cools to water at 0°C ?
- (c) How much energy is released or absorbed when 1 gm of water at 0°C freezes to ice at 0°C ?

(d) How much energy is released or absorbed when 1 gm of steam at 100°C turns to ice at 0°C ?

Ans: (a) We know that latent heat of vaporization of water (L) = 540 cal/g

Mass of water (m) = 1 g

$$\begin{aligned}\text{Energy transfer (Q)} &= mL \\ &= 1 \times 540 \\ &= 540\text{ cal.}\end{aligned}$$

Hence heat energy is transferred when 1 gm of boiling water at 100°C condenses to water at 100°C is 540 cal.

(b) Given: -

Mass, $m = 1\text{ gm}$

Initial temperature = 100°C

Final temperature = 0°C

Change in temperature, $\Delta T = 100^{\circ}\text{C}$

The specific heat capacity of water is, $S = 1\text{ cal/g}^{\circ}\text{C}$

We know that $Q = mS\Delta T$

Hence, energy transferred when 1g of boiling water at 100°C cools to water at 0°C -

$$\Rightarrow Q = mS\Delta T$$

$$\Rightarrow Q = 1 \times 1 \times 100$$

$$\Rightarrow Q = 100\text{ cal}$$

100 cal energy is transferred when 1 gm of boiling water at 100°C cools to water at 0°C

(c) Latent heat of ice = 80 cal

Mass of ice = 1 g

Heat energy Released $Q = mL$

$$\Rightarrow Q = 1 \times 80$$

$$= 80 \text{ cal}$$

80 cal energy is released or absorbed when 1 gm of water at 0°C freezes to ice at 0°C

(d) We know that

Latent Heat of steam (L_s) = 540 cal/g

Latent Heat of ice (L_i) = 80 cal/g

Specific Heat of water (s_w) = 1 cal/g°C

Mass of steam (m) = 1 g

There are three stages from formation of ice.

1). 100°C steam to 100°C water

Heat released $Q_1 = mL_s$

$$= 1 \times 540 = 540 \text{ cal}$$

2). 100°C water to 0°C of water

Heat energy released $Q_2 = ms(T_2 - T_1)$

$$= 1 \times 1 \times (100 - 0)$$

$$= 100$$

3). 0°C of water to 0°C of ice

Heat released $Q_3 = mL_i$

$$= 1 \times 80$$

$$=80$$

Total heat energy released

$$Q = Q_1 + Q_2 + Q_3$$

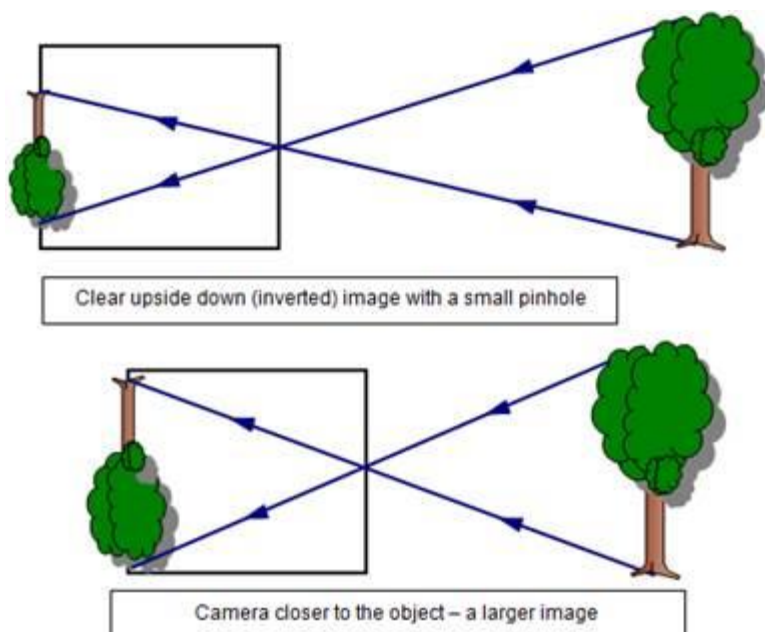
$$Q = 540 + 100 + 80$$

$$Q = 720 \text{ cal}$$

720 cal energy is released or absorbed when 1 gm of steam at 100°C turns to ice at 0°C.

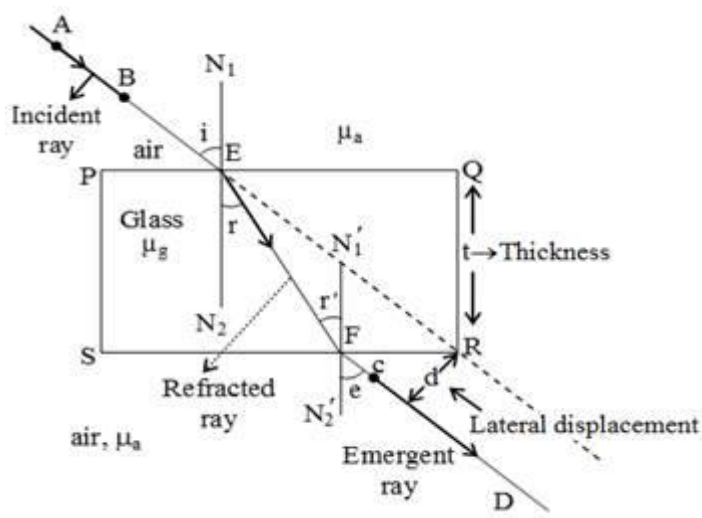
16. Draw and explain the process of formation of image with a Pinhole camera.

Ans: A pinhole camera is a simple camera without lens. It is made by taking a box in which is completely light proof that is dark with a small hole at one side. Light from the object enters this extremely small hole & projects an inverted image on the opposite side of the box. The size of the hole should be $1/100$ or less of the distance between it and the projected image. If the hole is too small then it would result in diffraction and if it's too large then we will see the object as it is. Pin hole camera works on the principle called rectilinear theory of light which states that light travels in a straight path.



1. The tree is being enlightened by sunlight. So the whole tree act as a source of light.
 2. Light from the top of the tree enters the hole and goes straight towards the bottom of the screen.
 3. Light from the bottom of the tree enters the hole and goes straight towards the top of the screen.
 4. So, in this way light coming from the tree which travels in a particular direction enters the pinhole & light which travels in other directions were blocked.
 5. This creates a sharp inverted image of the object.
17. Explain the refraction of light through a glass-slab with neat ray diagram.

Ans: Consider the rectangular slab as shown in figure.



AE = incident ray;

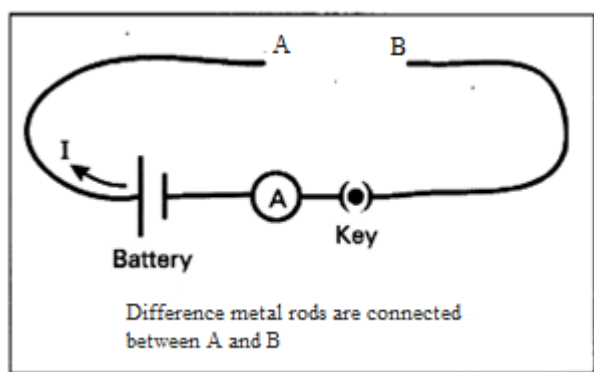
I = Angle of incidence;

R = Angle of refraction;

- On entering the glass slab the light ray bends towards the normal and travels along EF.
- The refracted ray EF is incident on the face SR at angle of incidence r' .
- The emergent ray FD bends away from normal at angle of refraction e .
- The emergent ray FD is parallel to the incident ray AE but it has been laterally displaced with respect to incident ray.

18. How do you verify that resistance of a conductor is proportional to the length of the conductor for constant cross-section area and temperature?

Ans:



Procedure:

Create a circuit leaving as shown in above image and insert Ammeter and Plug Key. Connect iron rod and see the value of current. With increase length there is a decrease in current.

So, $R \propto L$

We proved, resistance was more when the cross - section area was less.

Therefore , $R \propto L/A$

Thus we verify l and A

GROUP- B

19. How chemical displacement reactions differ from chemical decomposition reaction? Explain with an example for each.

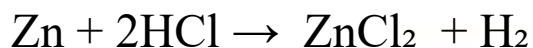
Ans: Decomposition reaction is single compound break into two or more simple substances under suitable conditions.

Calcium carbonate is decompose by heating and form calcium oxide and carbon dioxide



Displacement reaction in which a more reactive element displaces a less reactive element.

Zinc react with hydrochloric acid and zinc displace the hydrogen and produce zinc chloride and hydrogen gas.



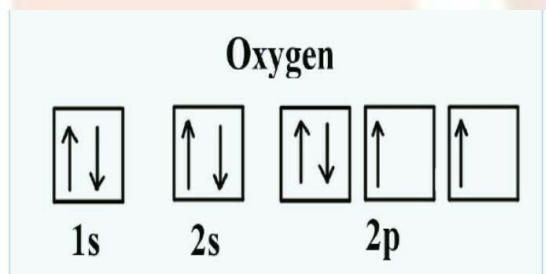
20. Explain Hund's rule with an example.

Ans: Hund's rule states that:

1. Every orbital in a sublevel is singly occupied before any orbital is doubly occupied.
2. All of the electrons in singly occupied orbitals have the same spin (to maximize total spin).

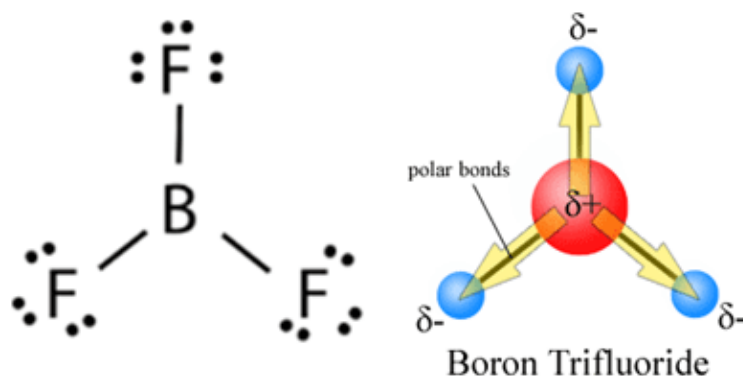
EXAMPLE: NITROGEN ATOMS

Oxygen ($Z = 8$) atom: $1s^2 2s^2 2p^4$



21. Explain the formation of the BF_3 molecule using hybridisation.

Ans:



BF_3 has a boron atom with three outer-shell electrons in its ground state and three fluorine atoms containing seven outer electrons. Further, if we observe closely, one boron electron is unpaired in the ground state. During the formation of this compound, the 2s orbital and two 2p orbitals hybridize. Only one of the empty p-orbital is left behind as the lone pair. In short, Boron needs 3 hybridised orbitals to make bonds with 3 atoms of F where the 2p_z orbitals get overlapped with these hybridised sp² orbitals and bonds are formed.

22. Suggest a test to find the hardness of water and explain its procedure.

Ans: We can take 2 buckets – One filled with distilled water and other with hard water. Add detergent to both the buckets and mix it well.

The bucket with hard water shows less foam due to presence of calcium or magnesium ions. So instead of foam there will be a white layer. Whereas bucket in distilled water will show lot of foam.

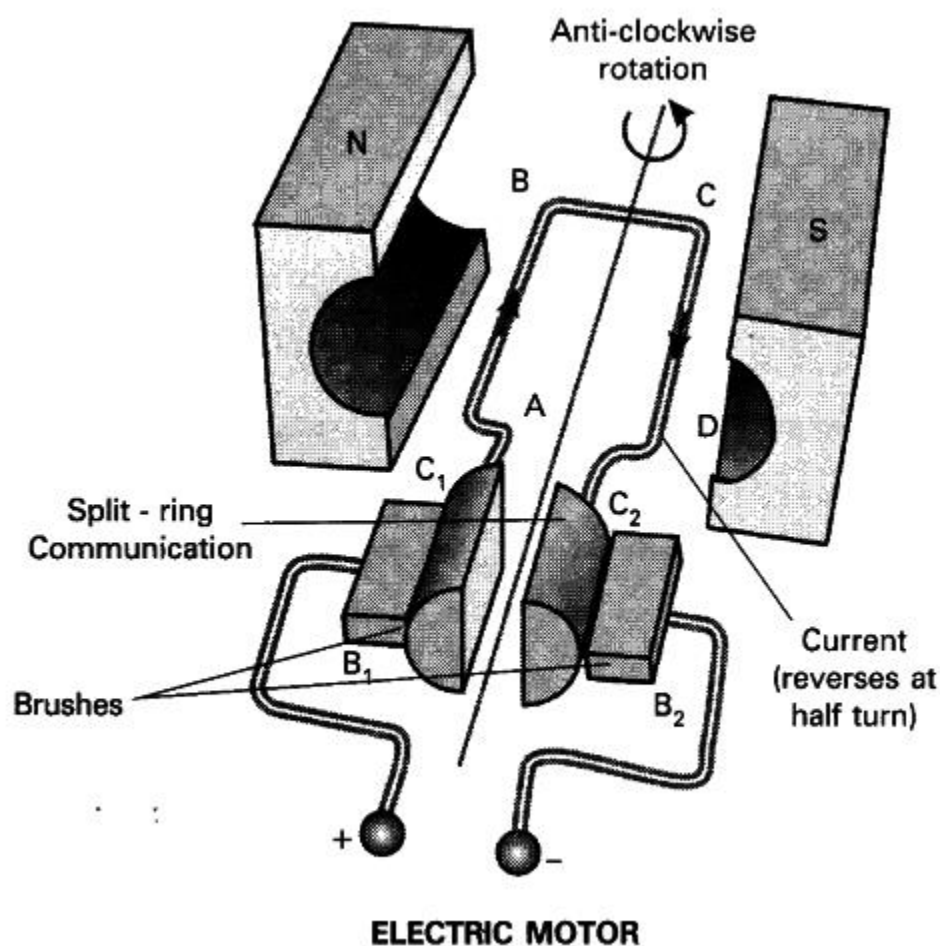
SECTION-IV

1x5=5

NOTE:

1. Answer ANY ONE question from the following.
2. This question carries FIVE marks.
23. Draw a neat diagram of Electric motor and name the parts.

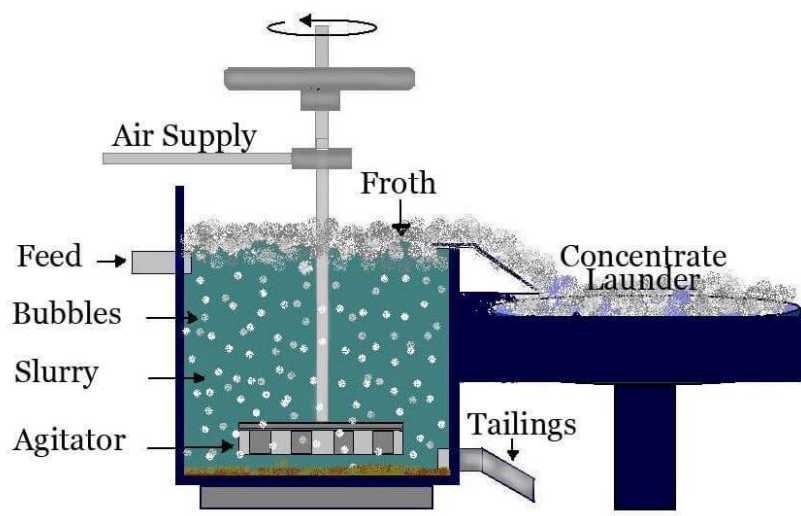
Ans:



24. Draw the diagram showing froth floatation method and label its parts.

Ans:

FLOTATION PROCESS



Part - B**Time: 30 minutes****Marks: 15**

- Note:**
1. Answer **all** the questions.
 2. Each question carries $\frac{1}{2}$ marks.
 3. Marks will **not** be awarded in case of any overwriting, re-writing or erased answers.

I. Write the CAPITAL LETTERS showing the correct answers for the following questions in the brackets provided against them.

$20 \times \frac{1}{2} = 10$

1. Boiling point of water at normal atmospheric pressure
..... []

(A) 0°C

(B) 100°C

(C) 110°C

(D) -4°C

Ans: At 1 atmosphere of pressure water boils at 100°C .

2. Magnification (m) =

[]

(A) v/u

(B) u/v

(C) h_o/h_i

(D) h_i/h_o

Ans - $M = H_i/H_o$. Where M stands for magnification, H_i is the height of the image, H_o is the height of the object.

3. If an object is placed at 'C' of a concave mirror, the position of the image is
[]

(A) at infinity.

(B) between F and C.

(C) at C.

(D) beyond C.

Ans: A real and inverted image will be formed at the same place i.e. at 'C'

4. The refractive index of glass with respect to air is 2.

Then the critical angle of glass-air interface is
[]

(A) 0°

(B) 45°

(C) 30°

(D) 60°

Ans: Relative refractive index = 2 and according to Snell's law,

$$\sin i / \sin r = 2 \text{ and } \sin i / \sin r = 1 / \sin c$$

$$2 = 1 / \sin c$$

$$\sin c = 1/2$$

$$\sin c = \sin 30$$

$$c = 30^\circ$$

Therefore Critical angle = 30°

5. Which one of the following materials cannot be used to make lens? []

- (A) Water
- (B) Glass
- (C) Plastic
- (D) Clay

Ans: Clay cannot be used to make lens since clay is not transparent and the light rays will not pass through it.

6. During refraction will not change. []

- (A) Wavelength
- (B) Frequency
- (C) Speed of light
- (D) All the above.

Ans: The frequency of the light will not change since it is the characteristics of the light and it is not dependent on the medium.

7. A charge is moved from point A to point B. The work done to move a unit charge during this process, is
[]

- (A) Potential at A.
- (B) Potential at B.
- (C) Current from A to B.
- (D) Potential difference between A and B.

Ans: Potential difference between A and B is the work done to move a unit charge.

8. A thick wire has a Resistance than thin wire.
[]

- | | |
|-----------|--------------|
| (A) More | (B) less |
| (C) equal | (D) A and B. |

Ans: A thick wire has less resistance than that of a thin wire.

9. Which converts mechanical energy into electrical energy?
[]

- | | |
|---------------|-------------|
| (A) Motor | (B) Battery |
| (C) Generator | (D) Switch |

Ans: A generator converts mechanical energy into electrical energy

10. The SI unit of magnetic field induction is
[]

- | | |
|-------------|-------------|
| (A) Tesla | (B) Weber |
| (C) Weber/m | (D) Weber.m |

Ans: Tesla is the SI unit of magnetic field induction and its symbol is T.

11. $C_6H_{12}O_6 \rightarrow C_2H_5OH + CO_2$ ischemical reaction.
[]

- | | |
|------------------|--------------------------|
| (A) Combination | (B) decomposition |
| (C) Displacement | (D) double decomposition |

Ans: Combustion reaction

12. Is used for treating indigestion.
[]

- | | |
|----------------|----------------|
| (A) antibiotic | (B) analgesic |
| (C) antacid | (D) antiseptic |

Ans: Antacid is used for treating indigestion because it helps in reducing stomach acid.

13. Colour of Methyl orange in alkali condition is
[]

- | | |
|------------|------------|
| (A) orange | (B) yellow |
| (C) red | (D) blue |

Ans: Methyl orange turns from red to orange and finally to yellow as the solution becomes less acidic.

14. The maximum number of electrons present in K shell are
[]

- | | |
|-------|-------|
| (A) 2 | (B) 4 |
| (C) 6 | (D) 8 |

Ans: The K shell contains only 2 electrons.

15. The value of Planck's constant is
[]

- | | |
|--|---------------------------------------|
| (A) $6.023 \times 10^{-34} \text{ Js}$ | (B) $6.626 \times 10^{34} \text{ Js}$ |
| (C) $6.626 \times 10^{-36} \text{ Js}$ | (D) None |

Ans: The Value of Planck's constant is $6.626 \times 10^{-36} \text{ Js}$.

16. Number of elements present in Period 1 are []

- [illegible]

Ans: Period 1 has only 2 elements Hydrogen and Helium

17. Which of the following elements is electronegative? []

- (A) Sodium (B) Oxygen
(C) Magnesium (D) Calcium

Ans: Oxygen is the electronegative element.

18. The bond angle in Methane []

- (A) $104^{\circ}31'$ (B) $107^{\circ}48'$
(C) 180° (D) $109^{\circ}28'$

Ans: Methane has a tetrahedral arrangement with an angle of $109^{\circ}28'$

19. The reducing agent in Thermite process is
[]

- (A) Al (B) Mg
(C) Fe (D) Si

Ans: Aluminium (Al) is used as the reducing agent in Thermite process.

20. Which one of the following hydrocarbons can show isomerism? []

(A) C_2H_4

(B) C_2H_6

(C) C_3H_8

(D) C_4H_{10}

Ans: C_4H_{10} Butane shows isomerism.

II. Fill in the blanks with suitable answers.

Each question carries $\frac{1}{2}$ marks.

$5 \times \frac{1}{2} = 2\frac{1}{2}$

21. The geometric centre of the mirror is Vertex.

22. At critical angle of incidence, the angle of refraction is 90°

23. The lens which can form real and virtual images is

Convex

24. The kilowatt hour is the unit ofEnergy.....

25. Faraday's law of induction is the consequence of ...lenz's law.....

III. Match the following. Each question carries $\frac{1}{2}$ marks.

$$5 \times \frac{1}{2} = 2\frac{1}{2}$$

Group - A**26.** Plaster of Paris**27.** Gypsum**28.** Bleaching powder**29.** Baking Soda**30.** Washing Soda

[D]

[G]

[E]

[B]

[A]

Group - B(A) Na_2CO_3 (B) NaHCO_3 (C) Na_2HCO_2 (D) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ (E) CaOCl_2 (F) $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ (G) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

